DEPTH IS UNCHANGED BY HENSELIZATION AND STRICT HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author's proof is terse but complete: it names the exact identification or cited result that settles the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- The henselization and strict henselization of a Noetherian local ring are Noetherian. Lemma \ref{lemma-henselization-noetherian}, cited at proof lines 11166-11167. Makes depth defined for the henselizations.
- Depth formula along the henselization maps. Algebra, Lemma \ref{algebra-lemma-apply-grothendieck}, cited at proof lines 11168-11170. Gives the equality of depths.

Source reference. Original lemma 06LL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8470–8470. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06LJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8468–8468. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0337: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4046–4046. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Given a Noetherian local ring R we have $depth(R) = depth(R^h) = depth(R^{sh})$.*

Proof. By Lemma 06LJ we know that R^h and R^{sh} are Noetherian. Hence the lemma follows from Algebra, Lemma 0337. □